9.A Graph Transformations of Rational Functions

Rational functions exhibit predictable structural behavior governed by algebraic transformations. Rather than relying on point-by-point plotting, effective analysis depends on recognizing how transformations alter the fundamental structure of a parent function.

By interpreting parameters within the function, we can determine asymptotes, orientation, and relative position with precision. This allows for efficient graphing and deeper conceptual understanding.

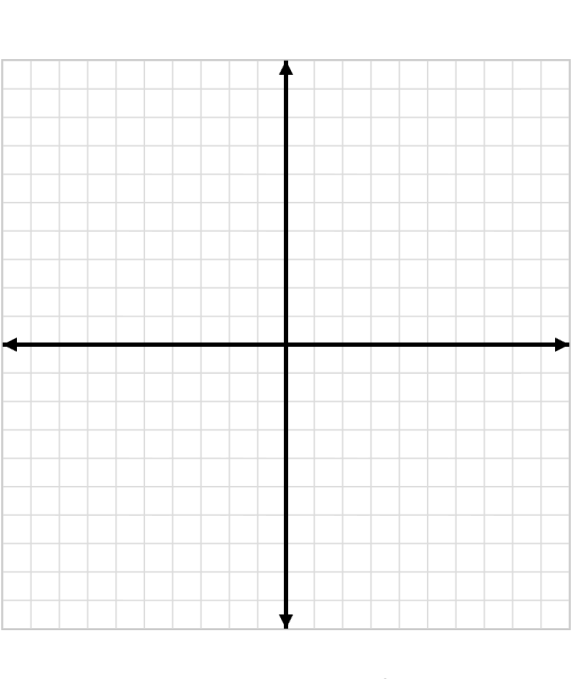
"Mathematics reveals structure — it does not create it."

Learning Targets

I can identify horizontal and vertical transformations of rational functions.
I can determine the effects of reflections and stretches/compressions.
I can sketch transformed rational functions using structural reasoning.

Introduction

The parent rational function is:
$f(x) = \frac{1}{x}$



This function is defined for all real values of x except $x = 0$. At this value, the function is undefined, creating a vertical asymptote. The graph also approaches, but never touches, the horizontal asymptote at $y = 0$.
Unlike polynomial functions, rational functions do not form continuous curves. Instead, they consist of separate branches that approach asymptotes. These asymptotes define the structure of the graph and guide how it behaves.

> **Key Idea:** Transformations do not change the fundamental structure of a rational function. They shift and reshape the graph while preserving its asymptotic behavior.

Connections

Prior Knowledge:
- Transformations of linear, quadratic, and exponential functions
- Understanding parent functions and how parameters affect graphs
- Graphing with symmetry, intercepts, and basic function behavior

Cross-Disciplinary Connections:
- Physics: inverse and inverse-square relationships (gravity, electric force)
- Chemistry: concentration vs reaction rate relationships
- Computer Science: asymptotic behavior and efficiency of algorithms

Real-World Applications:
- Cost per person as group size increases
- Time required to complete tasks with increasing resources
- Situations where values approach limits but never reach them

Key Concepts

General Form of a Transformed Rational Function

$f(x) = \frac{a}{x-h} + k$

Parameters and Their Effects:
- **a** → controls vertical stretch/compression and reflection
- **h** → shifts the graph left or right
- **k** → shifts the graph up or down

Asymptotes

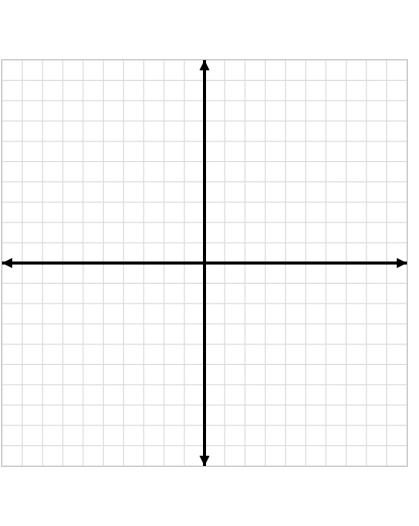
- **Vertical Asymptote:** $x = h$
- **Horizontal Asymptote:** $y = k$

Transformation Summary

- $x - h$ → shift right h units
- $x + h$ → shift left h units
- $+k$ → shift up k units
- $-k$ → shift down k units
- $a > 1$ → vertical stretch
- $0 < a < 1$ → vertical compression
- $a < 0$ → reflection across the x-axis

Structural Behavior

Rational functions based on $\frac{1}{x}$ always have two branches. These branches are separated by the vertical asymptote and approach both asymptotes without ever touching them.



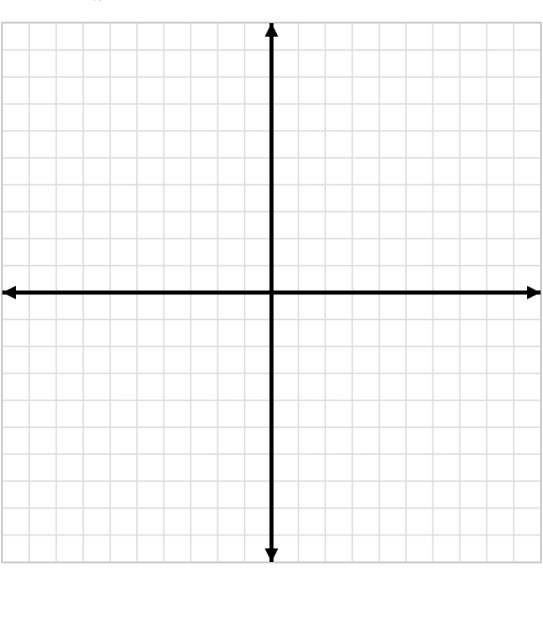
Strategies

Graphing Strategy for Rational Functions:
1. **Identify asymptotes first.**
   Determine the vertical asymptote from the denominator and the horizontal asymptote from the constant shift.
2. **Analyze transformations.**
   Identify shifts (h, k), reflections, and stretches using the parameters in the function.
3. **Determine branch orientation.**
   Use the sign of a to determine which quadrants the graph will occupy relative to the asymptotes.
4. **Sketch relative to asymptotes.**
   Draw branches approaching asymptotes without touching them. Focus on structure rather than plotting many points.
5. **Verify with key points (if needed).**
   Use one or two points to confirm accuracy, not to build the entire graph.

Examples

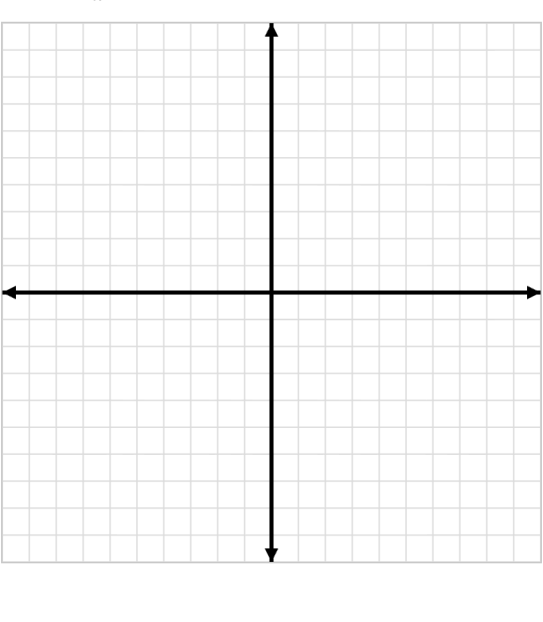
Example 1 — Horizontal Shift

Identify the vertical asymptote before sketching.
$f(x) = \frac{1}{x-2}$



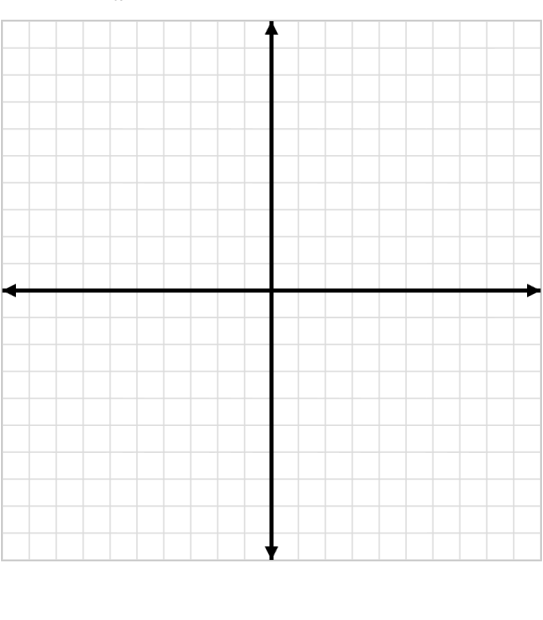
Example 2 — Vertical Translation

Determine the horizontal and vertical shifts.
$f(x) = \frac{1}{x} + 3$



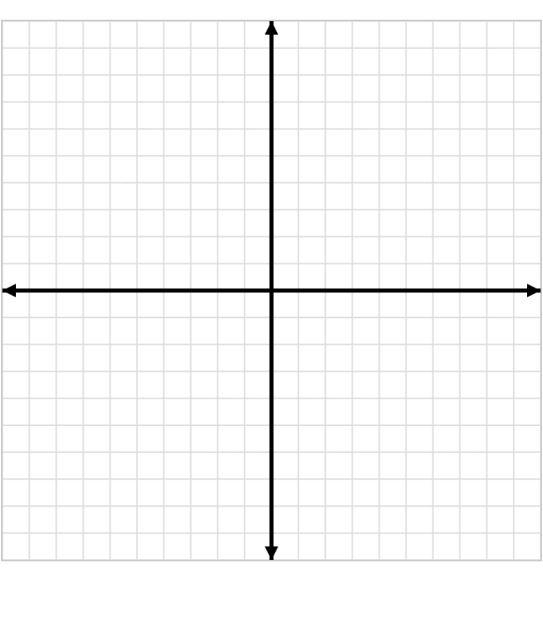
Example 3 — Reflection Across the x-axis

How does the negative sign affect orientation?
$f(x) = -\frac{1}{x}$



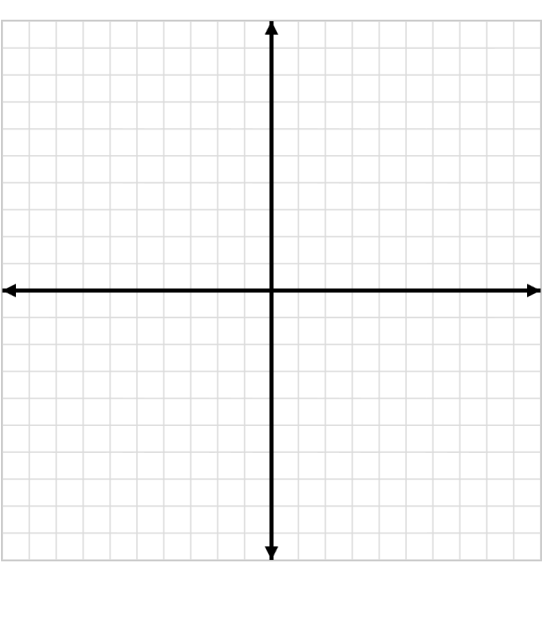
Example 4 — Vertical Stretch

Describe how the coefficient changes the graph.
$f(x) = \frac{2}{x}$



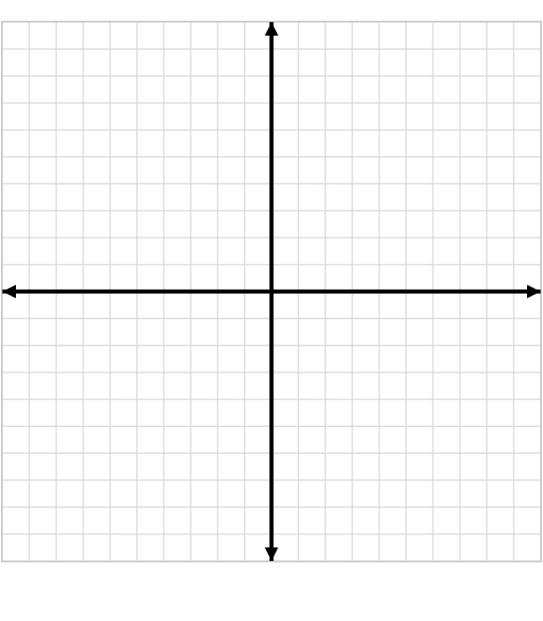
Example 5 — Combined Transformations

Identify both asymptotes and describe all transformations.
$f(x) = \frac{1}{x+4} - 1$



Example 6 — Reflection and Translation

Determine asymptotes, orientation, and transformations.
$f(x) = -\frac{3}{x-1} + 2$



Common Mistakes

- **Ignoring asymptotes:**
  Students often try to graph the function without first identifying vertical and horizontal asymptotes, which leads to incorrect structure.
- **Reversing horizontal shifts:**
  Remember that $x - h$ shifts right and $x + h$ shifts left. This is opposite of what many expect.
- **Forgetting reflections:**
  A negative value of a reflects the graph across the x-axis, changing the orientation of the branches.
- **Over-relying on plotting points:**
  Rational functions should be sketched using asymptotes and structure, not built point-by-point.
- **Drawing branches that cross asymptotes:**
  Graphs should approach asymptotes but never touch or cross them.

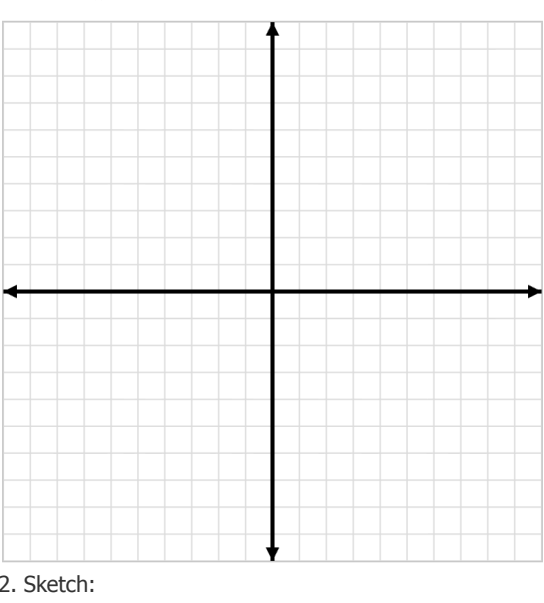
Summary

- Rational functions are transformations of the parent function $frac1x$.
- Asymptotes define the structure of the graph.
- The parameters a, h, and k control reflection, stretch, and translation.
- Graphing should begin with identifying asymptotes before sketching.
- Understanding structure allows you to graph efficiently without relying on tables of values.

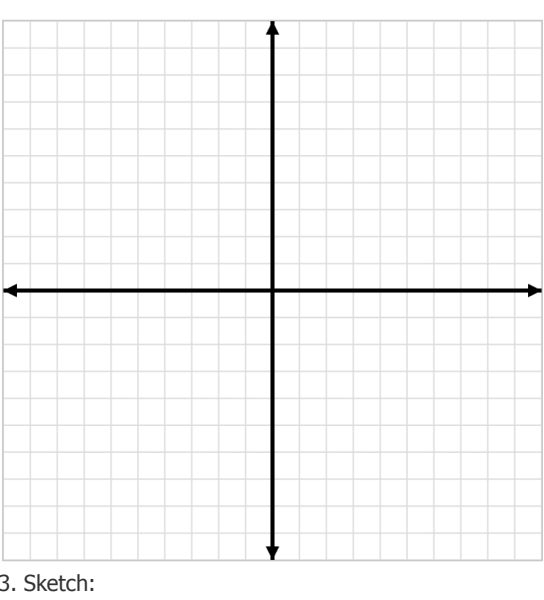
Practice Set

Level 1 — Easy

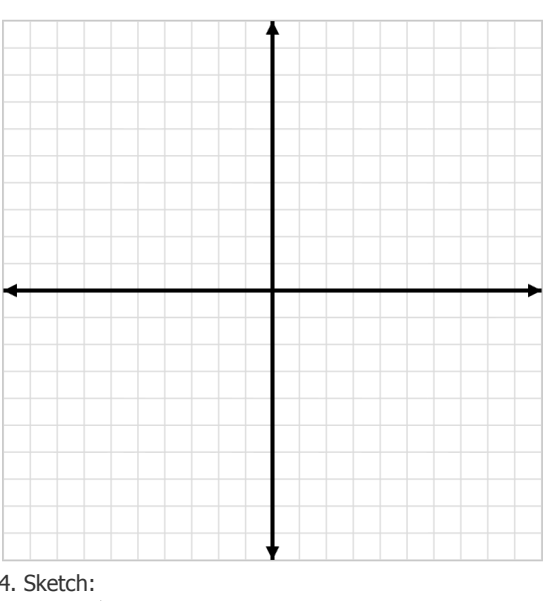
1. Sketch:
$f(x) = \frac{1}{x-1}$



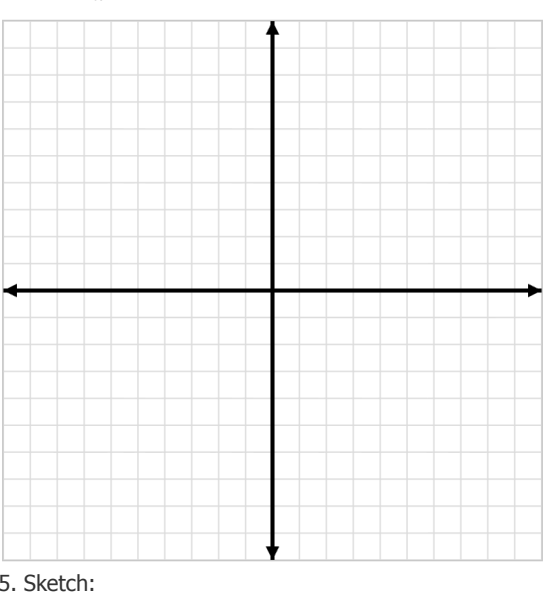
2. Sketch:
$f(x) = \frac{1}{x} + 2$



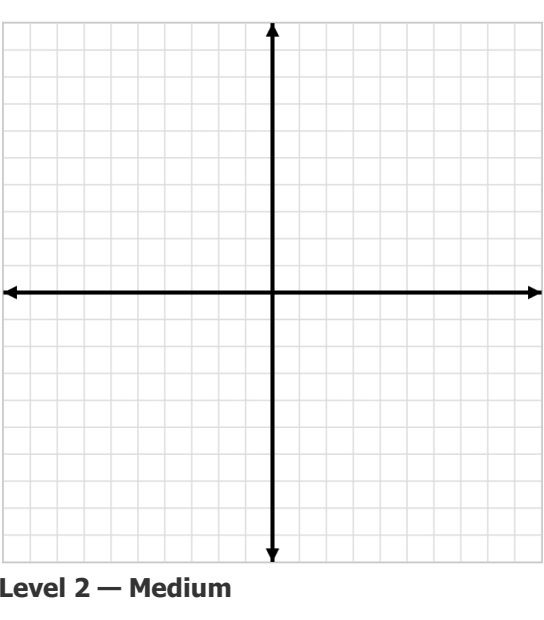
3. Sketch:
$f(x) = \frac{1}{x+3}$



4. Sketch:
$f(x) = \frac{1}{x} - 4$

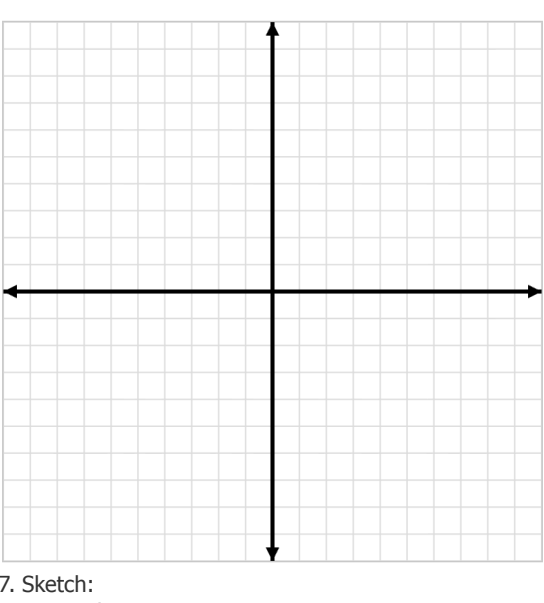


5. Sketch:
$f(x) = \frac{1}{x-5}$

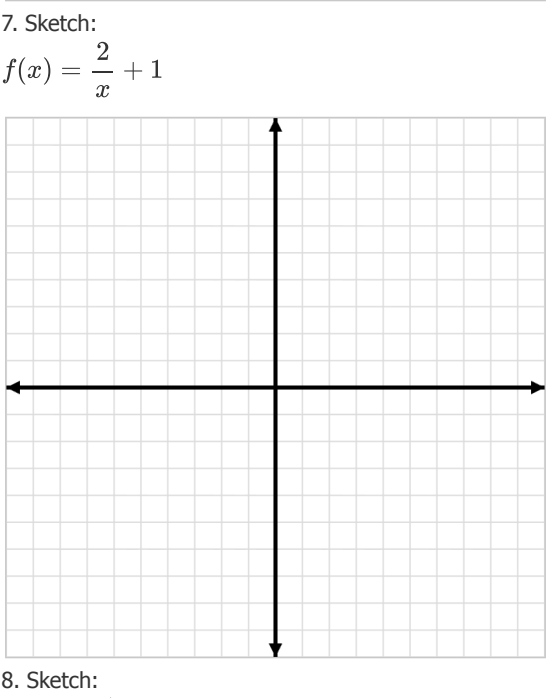


Level 2 — Medium

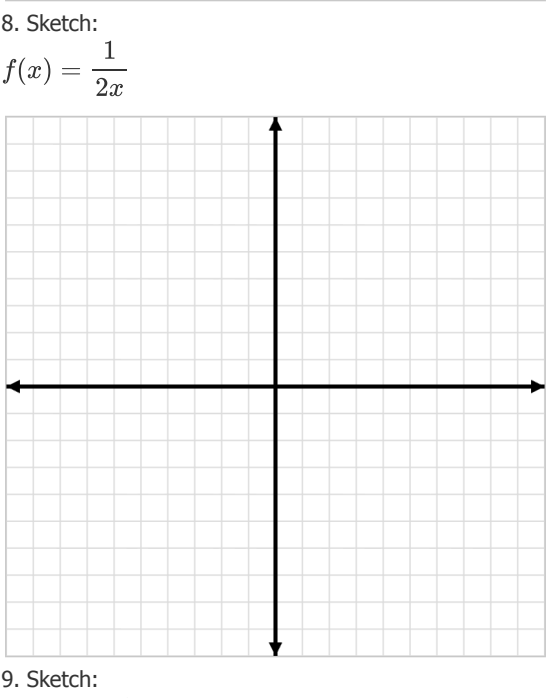
6. Sketch:
$f(x) = -\frac{1}{x-2}$



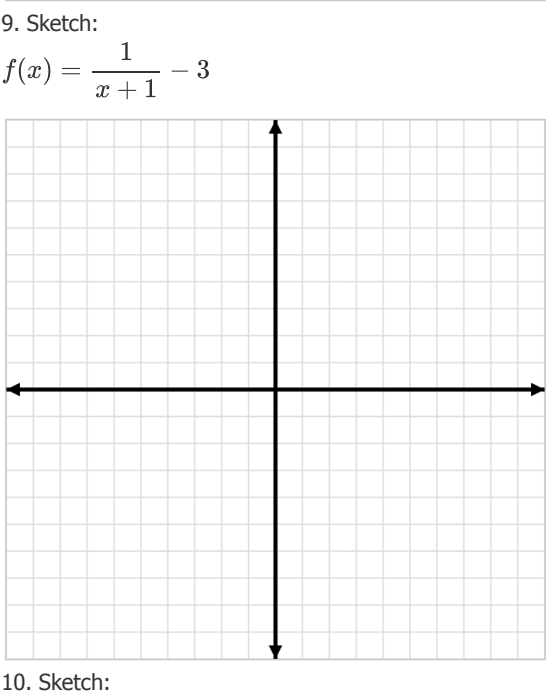
7. Sketch:
$f(x) = \frac{2}{x} + 1$



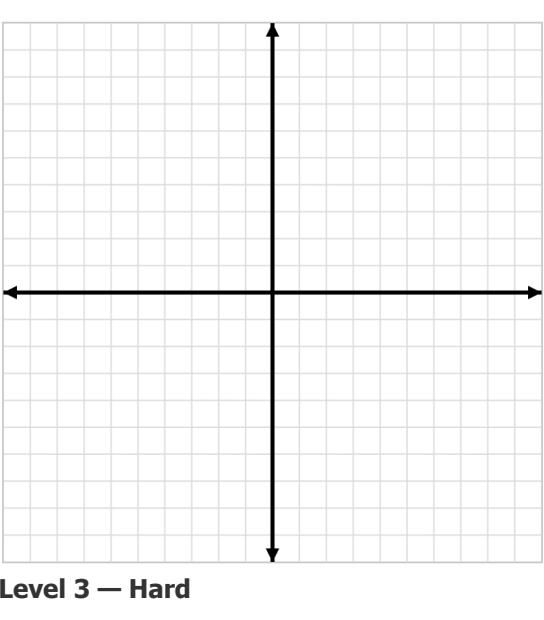
8. Sketch:
$f(x) = \frac{1}{2x}$



9. Sketch:
$f(x) = \frac{1}{x+1} - 3$

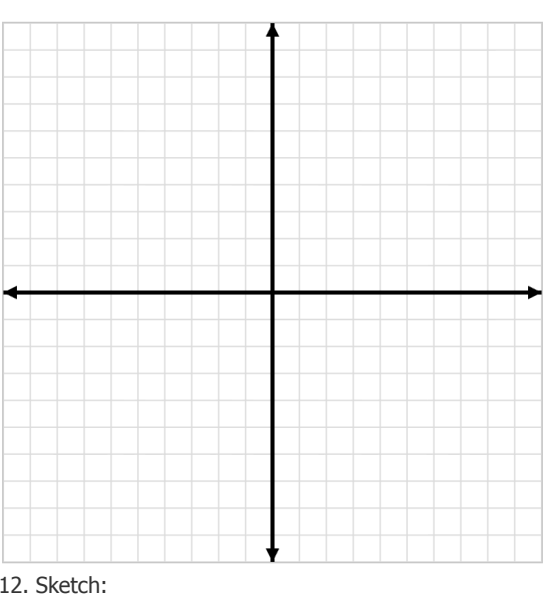


10. Sketch:
$f(x) = -\frac{2}{x-4}$

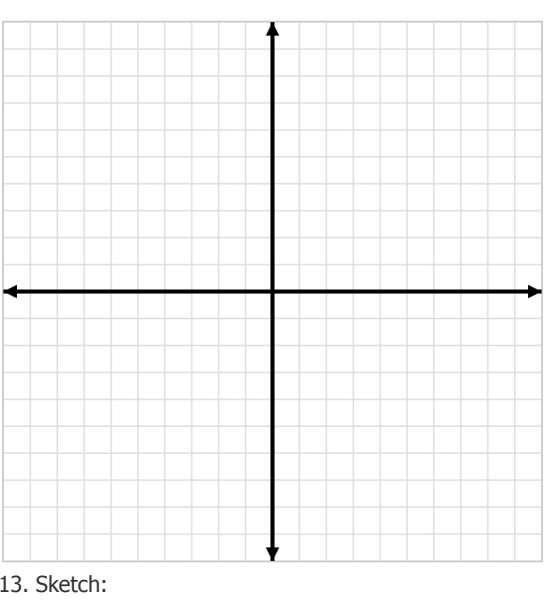


Level 3 — Hard

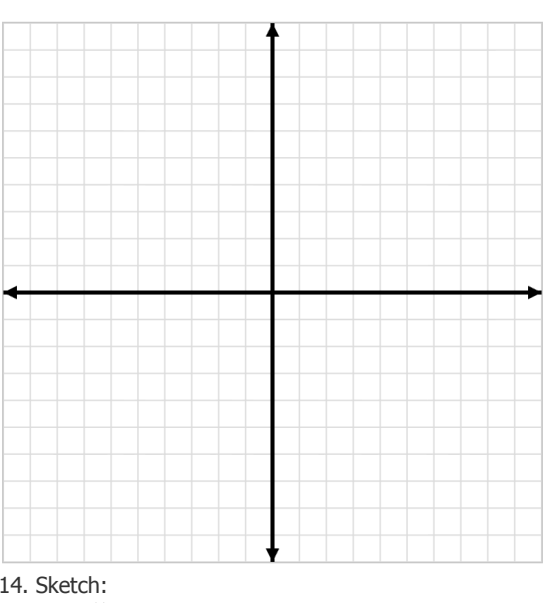
11. Sketch:
$f(x) = \frac{3}{x-2} + 1$



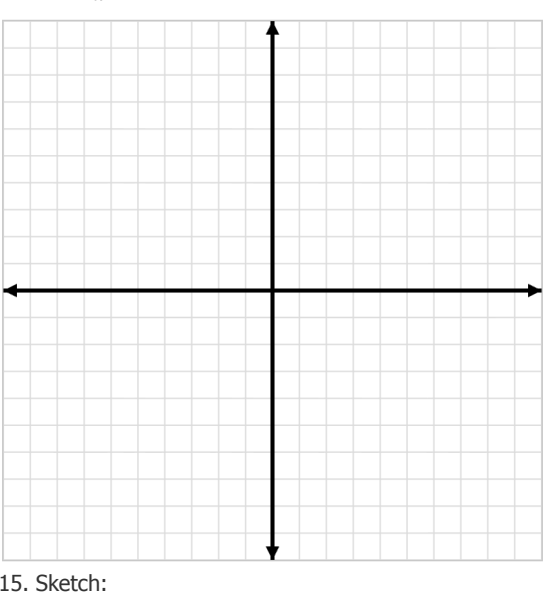
12. Sketch:
$f(x) = -\frac{4}{x+3} - 2$



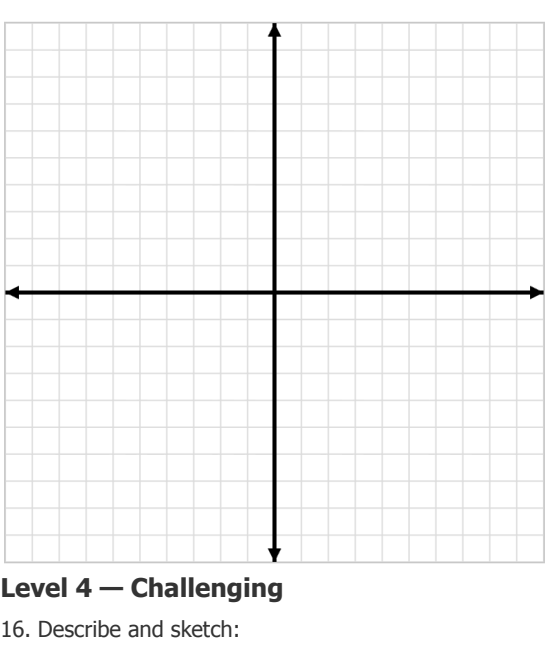
13. Sketch:
$f(x) = \frac{1}{x-1} + 4$



14. Sketch:
$f(x) = \frac{5}{x} + 2$

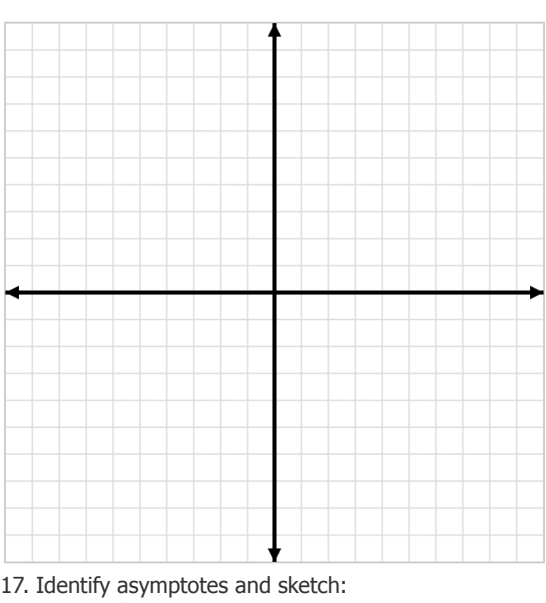


15. Sketch:
$f(x) = -\frac{1}{x-3} + 1$

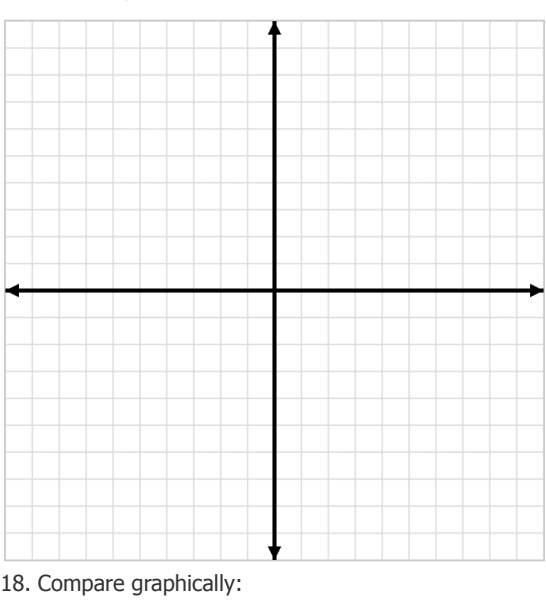


Level 4 — Challenging

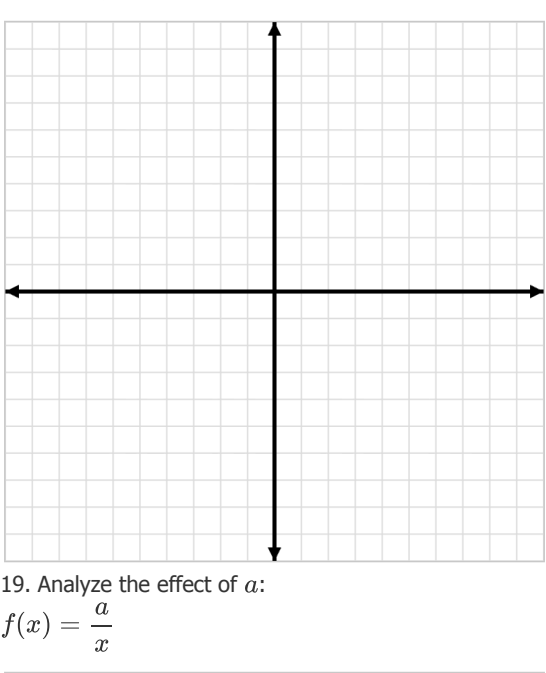
16. Describe and sketch:
$f(x) = \frac{-2}{x-3} + 4$



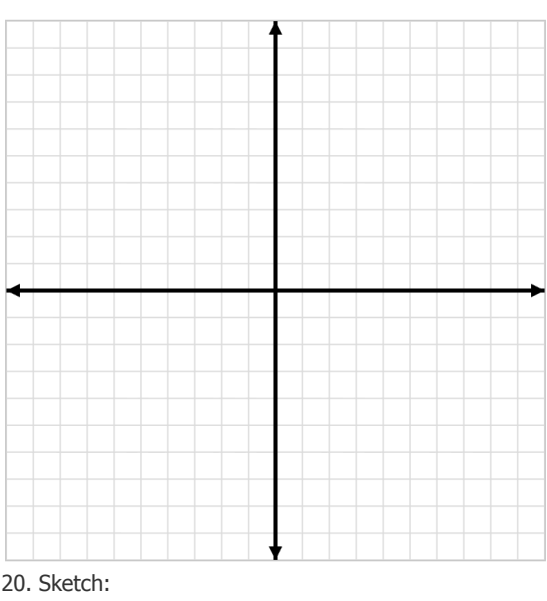
17. Identify asymptotes and sketch:
$f(x) = \frac{1}{x+2} - 2$



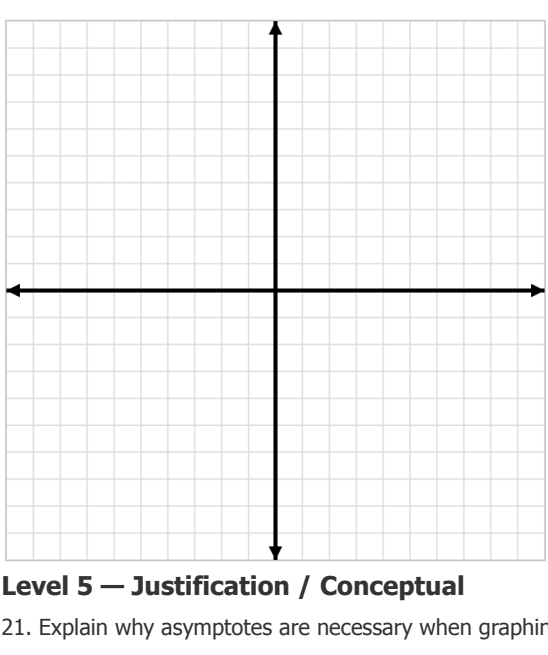
18. Compare graphically:
$f(x) = \frac{1}{x-4}$



19. Analyze the effect of a:
$f(x) = \frac{a}{x}$



20. Sketch:
$f(x) = \frac{1}{2(x-1)} + 3$



Level 5 — Justification / Conceptual

21. Explain why asymptotes are necessary when graphing rational functions.
22. Explain how a negative coefficient changes the graph of $frac1x$.
23. Justify why $x - h$ shifts the graph to the right.
24. Explain why rational functions have two branches.
25. Describe how transformations allow graphing without plotting many points.

Answer Key

Level 1 — Easy

1. VA: $x = 1$, HA: $y = 0$
2. VA: $x = 0$, HA: $y = 2$
3. VA: $x = -3$, HA: $y = 0$
4. VA: $x = 0$, HA: $y = -4$
5. VA: $x = 5$, HA: $y = 0$

Level 2 — Medium

6. VA: $x = 2$, HA: $y = 0$, reflection
7. VA: $x = 0$, HA: $y = 1$, stretch (factor 2)
8. VA: $x = 0$, HA: $y = 0$, compression
9. VA: $x = -1$, HA: $y = -3$
10. VA: $x = 4$, HA: $y = 0$, reflection, stretch

Level 3 — Hard

11. VA: $x = 2$, HA: $y = 1$, stretch (factor 3)
12. VA: $x = -3$, HA: $y = -2$, reflection, stretch
13. VA: $x = 1$, HA: $y = 4$
14. VA: $x = 0$, HA: $y = 2$, stretch (factor 5)
15. VA: $x = 3$, HA: $y = 1$, reflection

Level 4 — Challenging

16. VA: $x = 3$, HA: $y = 4$, reflection, stretch (factor 2)
17. VA: $x = -2$, HA: $y = -2$
18. Shift right 4 units
19. a controls stretch/compression and reflection
20. VA: $x = 1$, HA: $y = 3$, compression

Level 5 — Justification / Conceptual

21. Asymptotes define the structure and behavior of the graph.
22. A negative coefficient reflects the graph across the x-axis, changing branch orientation.
23. Because the input must equal h to make the denominator zero, shifting the vertical asymptote right.
24. The function is undefined at the vertical asymptote, splitting the graph into two branches.
25. Transformations allow prediction of graph shape using structure instead of plotting points.

Detailed Example Solutions

Example 1 — Horizontal Shift

Given:
$f(x) = \frac{1}{x-2}$
Step 1: Identify asymptotes
Vertical asymptote occurs when the denominator is zero:
$x - 2 = 0 \Rightarrow x = 2$
Horizontal asymptote:
$y = 0$
Step 2: Describe transformation
The graph is shifted right 2 units from $frac1x$.
Final Answer:
VA: $x = 2$, HA: $y = 0$

Example 2 — Vertical Translation

Given:
$f(x) = \frac{1}{x} + 3$
Step 1: Identify asymptotes
Vertical asymptote:
$x = 0$
Horizontal asymptote:
$y = 3$
Step 2: Describe transformation
The graph is shifted up 3 units.
Final Answer:
VA: $x = 0$, HA: $y = 3$

Example 3 — Reflection Across the x-axis

Given:
$f(x) = -\frac{1}{x}$
Step 1: Identify asymptotes
Vertical asymptote:
$x = 0$
Horizontal asymptote:
$y = 0$
Step 2: Describe transformation
The negative sign reflects the graph across the x-axis.
Final Answer:
VA: $x = 0$, HA: $y = 0$, reflection across x-axis

Example 4 — Vertical Stretch

Given:
$f(x) = \frac{2}{x}$
Step 1: Identify asymptotes
Vertical asymptote:
$x = 0$
Horizontal asymptote:
$y = 0$
Step 2: Describe transformation
The factor of 2 creates a vertical stretch.
Final Answer:
VA: $x = 0$, HA: $y = 0$, stretch factor 2

Example 5 — Combined Transformations

Given:
$f(x) = \frac{1}{x+4} - 1$
Step 1: Identify asymptotes
Vertical asymptote:
$x + 4 = 0 \Rightarrow x = -4$
Horizontal asymptote:
$y = -1$
Step 2: Describe transformation
The graph shifts left 4 units and down 1 unit.
Final Answer:
VA: $x = -4$, HA: $y = -1$

Example 6 — Reflection and Translation

Given:
$f(x) = -\frac{3}{x-1} + 2$
Step 1: Identify asymptotes
Vertical asymptote:
$x - 1 = 0 \Rightarrow x = 1$
Horizontal asymptote:
$y = 2$
Step 2: Describe transformation
The graph is shifted right 1 unit and up 2 units.
The negative sign reflects the graph across the x-axis.
The factor of 3 creates a vertical stretch.
Final Answer:
VA: $x = 1$, HA: $y = 2$, reflection, stretch factor 3